TUPLE BASED COURSE REGISTRATION VALIDATION

Python Project Report

1.Overview

The Tuple Based Course Registration Validation system is designed to validate student course registrations using Python tuples. A tuple is used to store course details such as course ID, course name, course credits, and course capacity.

The system checks whether a selected course exists, whether the student satisfies the registration requirements, and whether the course has available seats. It displays appropriate messages for valid and invalid registrations.

The main objective of this project is to simplify course registration, reduce errors, and ensure that students register for courses according to predefined rules.

2.Example Policy Used

The following sample policies are used for course registration validation.

1. Each course must have a unique course ID.
2. Course details are stored using Python tuples.
3. Students must select a course from the available course list.
4. A student must have completed the required prerequisites, if applicable.
5. A student cannot register for a course if the registration limit has been reached.
6. The selected course ID must be valid.
7. Invalid course selections must be rejected.
8. The system must display a confirmation message for successful registration.

Example:

Course Details: ("CS101", "Python Programming", 4, 30)

- Course ID: CS101
- Course Name: Python Programming
- Credits: 4
- Maximum Seats: 30

If a student selects CS101 and the course has available seats, the registration is accepted, provided all other requirements are satisfied.

3.Problem Statements

1. Manual course registration may lead to errors in course selection.
2. Students may attempt to register for courses that do not exist.
3. Course registration must follow predefined eligibility rules.
4. Limited course seats must be managed properly.
5. Invalid course IDs may cause registration errors.
6. There is a need for a simple system to validate course registration automatically.

The proposed system uses Python tuples to store course information and validate student registration requests efficiently.

4.Objectives

1. To implement course registration validation using Python tuples.
2. To store course details in an organized format.
3. To verify whether a selected course exists.
4. To validate course IDs entered by students.
5. To check course availability and seat limits.
6. To reduce errors during course registration.
7. To display appropriate success and error messages.
8. To provide a simple and efficient course registration process.

5.Assumptions

1. Each course has a unique course ID.
2. Course details are stored in Python tuples.
3. The available course list is predefined.
4. Students enter valid registration information.
5. Course credit values are predefined.
6. Course capacity is specified for each course.
7. Only courses available in the system can be selected.
8. Registration is accepted only when all applicable validation rules are satisfied.
9. The sample system uses predefined course information.
10. The system displays an error message when registration validation fails.

6.Algorithm

Step 1: Start.

Step 2: Store course details in Python tuples.

Step 3: Enter the student ID and selected course ID.

Step 4: Search for the selected course in the available course list.

Step 5: If the course ID does not exist, display "Invalid Course ID."

Step 6: Otherwise, check the course registration requirements.

Step 7: Verify course availability and seat limits.

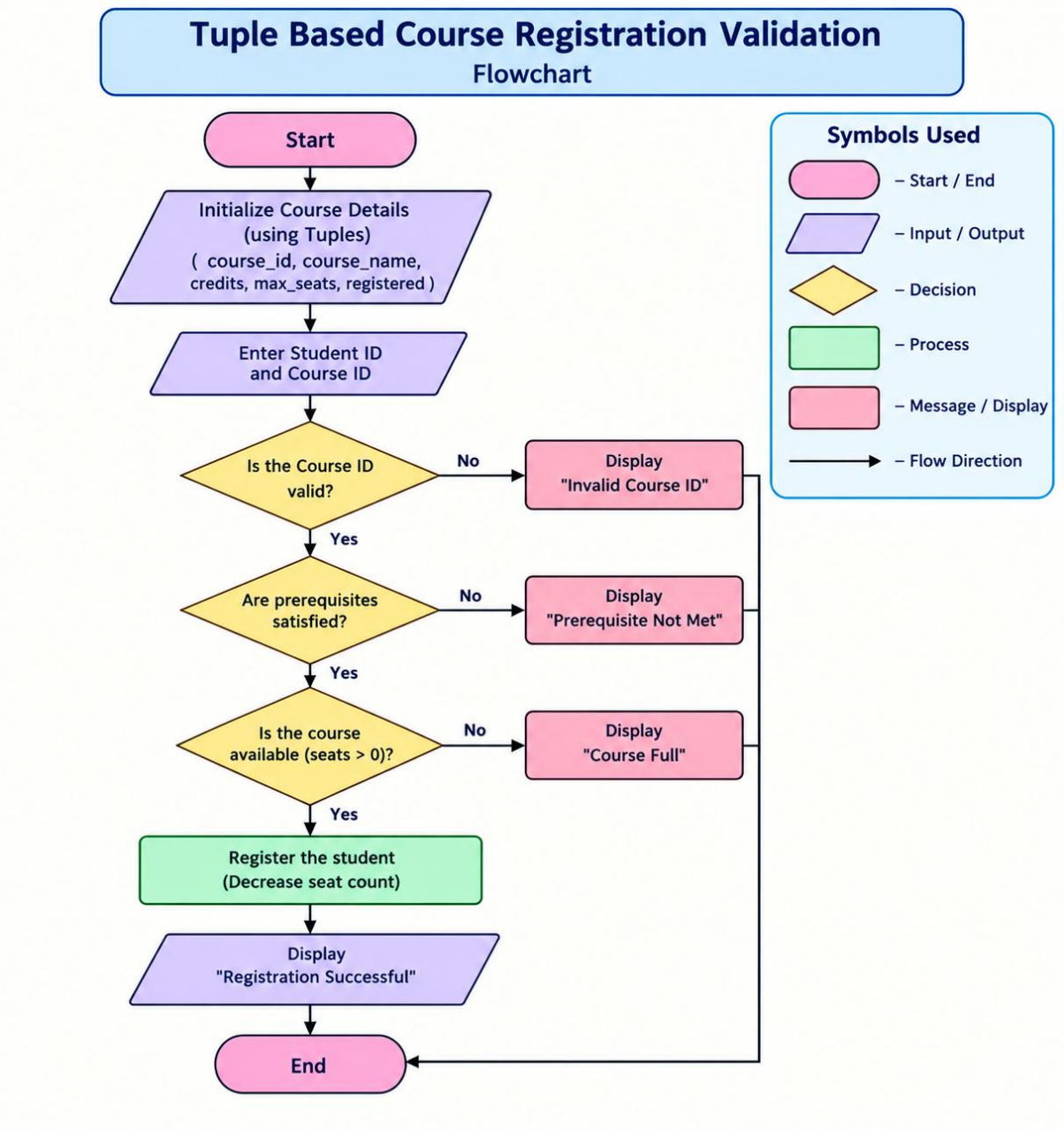
Step 8: If all conditions are satisfied, accept the registration.

Step 9: Otherwise, reject the registration and display the appropriate message.

Step 10: Display the registration status.

Step 11: Stop.

7.Flowchart



8.Pseudocode

START

STORE course details in TUPLES

INPUT Student_ID

INPUT Course_ID

IF Course_ID is NOT available

PRINT "Invalid Course ID"

ELSE IF Prerequisites are NOT satisfied

PRINT "Prerequisite Not Met"

ELSE IF Available_Seats = 0

PRINT "Course Full"

ELSE

REGISTER Student

REDUCE Available_Seats by 1

PRINT "Registration Successful"

ENDIF

STOP

9.Source Code

```
# Tuple Based Course Registration Validation

print("========================================")
print(" TUPLE BASED COURSE REGISTRATION")
print("========================================")

# Course details stored as tuples
courses = (
    ("CS101", "Python Programming", 4, 30, 1),
    ("CS102", "Java Programming", 3, 25, 0),
```

```
    ("CS103", "Database Systems", 4, 0, 1),
    ("CS104", "Web Development", 3, 20, 1)
)

# Tuple format:
# (Course_ID, Course_Name, Credits, Available_Seats, Prerequisite_Status)

student_id = input("Enter Student ID: ")
course_id = input("Enter Course ID: ").strip().upper()

# Search for the selected course
selected_course = None

for course in courses:
    if course[0] == course_id:
        selected_course = course
        break

# Validate course registration
if selected_course is None:
    print("\nInvalid Course ID!")

else:
    course_name = selected_course[1]
    credits = selected_course[2]
    seats = selected_course[3]
    prerequisite = selected_course[4]
```

```
    print("\nStudent ID:", student_id)
    print("Course Name:", course_name)
    print("Course Credits:", credits)

    if prerequisite == 0:
        print("Registration Failed: Prerequisite Not Met")

    elif seats == 0:
        print("Registration Failed: Course Full")

    else:
        print("Registration Successful!")
        print("Course ID:", course_id)
        print("Available Seats Before Registration:", seats)
        print("Available Seats After Registration:", seats - 1)

print("\nProgram Completed.")
```

10.Test Cases

Test Case ID	Student ID	Course ID	Condition	Expected Output
TC01	S101	CS101	Valid course, seats available, prerequisite satisfied	Registration Successful
TC02	S102	CS999	Invalid course ID	Invalid Course ID
TC03	S103	CS102	Prerequisite not satisfied	Registration Failed: Prerequisite Not Met
TC04	S104	CS103	No seats available	Registration Failed: Course Full
TC05	S105	CS104	Valid course, seats available	Registration Successful

Test Case ID	Student ID	Course ID	Condition	Expected Output
TC06	S106	cs101	Lowercase course ID entered	Registration Successful
TC07	S107	CS101	Course has 30 available seats	Registration Successful
TC08	S108	CS103	Course is full	Registration Failed: Course Full

11.Output Screenshots

```
========================================
 TUPLE BASED COURSE REGISTRATION
========================================
Enter Student ID: S101
Enter Course ID: CS101

Student ID: S101
Course Name: Python Programming
Course Credits: 4
Registration Successful!
Course ID: CS101
Available Seats Before Registration: 30
Available Seats After Registration: 29

Program Completed.

================ RESTART: C:\Users\VARSHINI\Desktop\python\activity2.py ============
========================================
 TUPLE BASED COURSE REGISTRATION
========================================
Enter Student ID: S102
Enter Course ID: CS999

Invalid Course ID!

Program Completed.
```

```
================ RESTART: C:\Users\VARSHINI\Desktop\python\activity2.py ==============
========================================
 TUPLE BASED COURSE REGISTRATION
========================================
Enter Student ID: S103
Enter Course ID: Cs103

Student ID: S103
Course Name: Database Systems
Course Credits: 4
Registration Failed: Course Full

Program Completed.

================ RESTART: C:\Users\VARSHINI\Desktop\python\activity2.py ==============
========================================
 TUPLE BASED COURSE REGISTRATION
========================================
Enter Student ID: S104
Enter Course ID: CS102

Student ID: S104
Course Name: Java Programming
Course Credits: 3
Registration Failed: Prerequisite Not Met

Program Completed.
```

12.Conclusion

The Tuple Based Course Registration Validation system provides a simple and efficient method for validating student course registrations using Python tuples. It checks course IDs, prerequisite requirements, and seat availability before confirming registration. The system reduces manual errors, improves accuracy, and displays appropriate messages for successful and unsuccessful registrations. It also demonstrates the practical use of tuples in storing and accessing course information. Thus, this project provides a basic and reliable solution for course registration validation.